Results and analysis

Introduction

This project involved using various models, including one sequence-to-sequence model and one autoregressive model for the task of machine translation. The use of machine translation models has become widespread. Given this growing use, it has become essential to evaluate the quality of these translations using various tools to ensure their reliability. Human translation is considered the most reliable and is generally used as the gold standard for evaluating translations produced by tools. Evaluation can be human or automatic. Human evaluation involves the analysis, by humans, of the translation outputs produced by tools using metrics such as adequacy (preservation of the meaning of the original sentence in the translation) and fluency (the phrasing in the target language: whether the sentence could be produced by a native speaker). However, human evaluation is extremely time-consuming and requires significant human resources, especially since these evaluations are not consistent and depend heavily on subjectivity. Automated evaluation, on the other hand, has the advantage of being more consistent because it is based on established metrics. Several metrics are used. The first is based on edit distance and involve measuring the distance between the reference translation and the candidate translation by calculating the number of operations required to transform one into the other. This includes metrics such as Word Error Rate (WER) and Translation Edit Rate (TER). The second measure is based on precision and recall and calculates the precision and recall between the n-grams of the reference translation and the proposed translation. Some examples include BLEU (Bilingual Evaluation Understudy) and METEOR. Finally, the metric based on semantic similarity uses embeddings to measure the semantic proximity between the reference translation and the proposed translation (BERTScore, COMET). For this project, we used three models: two sequence-to-sequence models, Helsinki-NLP/opus-mt-en-fr and google/flan-t5-large, and one autoregressive model, openai/gpt-oss120b:free.

Methodology and results

For this exercise, we used a file, sents.csv, which contains several sentences in English along with their French translations. The French sentences are used as the reference translations and serve as the gold standard for evaluating the translations produced by the systems. We randomly selected 20 sentences in French and English. The three different models (Helsinki-NLP/opus-mt-en-fr, google/flan-t5-large, and openai/gpt-oss-120b:free) are fed these 20 sentences and produce translations. HelsinkiNLP/opus-mt-en-fr is a model available on the Hugging Face library. It is a Transformer model specialized for translations from English to French, trained on the

OPUS corpus, a database containing several parallel corpora. The second model, google/flan-t5- large, is a T5-based model but fine-tuned for various NLP tasks.

We used it with the instruction: “translate from English to French.” In addition, we specified the maximum length, setting it to 100, to ensure that the translations would not be truncated. Finally, for the autoregressive model, we used openai/gpt-oss-120b:free, a free GPT model provided by OpenAI. We accessed the model using an API key obtained from OpenRouter. It is a free GPT model with 120 billion parameters. We provided the model with the following prompt:

Translate the following text from english to french. For each translation, respond in the form of a valid json object following this format: There should be no text before or after, simply follow this format.

{{"english": "...",

"french": "..."}}

The sentences are from diverse domains: astronomy, chemistry, daily life and others:: (text)

We placed particular emphasis on the output format because, after several trials, we noticed that the LLM’s output format varied greatly, and that it sometimes produced outputs that were difficult to use. We also specified that the texts covered different domains. With this prompt, the model outputs the expected format, from which we extract only the translations, in order to compare them to the reference translations and calculate the scores. We use three evaluation metrics for each model. First, we use Word Error Rate (WER). This is a measure based on edit distance that calculates the number of changes (deletions, insertions, substitutions) required to go from the translation proposed by the model to the reference translation (and vice versa). The changes are then divided by the total number of words in the reference translation. We chose this metric because it allows us to compare the proposed translation to the reference translation on a typographical level. The WER score must be between 0 and 1. The lower the score, the better the translation, as this implies that few changes are needed to move from the proposed translation to the reference. Thus, the proposed translation is close to the reference and is therefore a good translation. We used the jiwer package to calculate the WER. We also used BLEU (Bilingual Evaluation Understudy), which measures the difference between the proposed translation and the reference in terms of n-grams. Accuracy with BLEU is calculated by measuring the number of matching n-grams in the reference translation divided by the total number of n-grams in the reference. We access BLEU via the sacrebleu library and set the maximum n-gram size to 4. Finally, we used BERTScore, which evaluates translations based on semantic similarity. This model is based on lexical embeddings of the proposed translations and

the reference translations and measures the cosine similarity between the two to assign a score to the translation. For BERTScore, we calculate the precision of each translation.

	HelsinkiNLP/opus-mt-enfr	openai/gpt-oss120b:free	google/flan-t5-large
WER (unnormalized)	0.49	0.45	0.71
BLEU (unnormalized)	39.91%	47.98%	20.83%
Mean BERTScore	0.90	0.91	0.86

Unnormalized scores for the diverse models

	HelsinkiNLP/opus-mt-enfr	openai/gpt-oss120b:free	google/flan-t5-large
WER (normalized)	0.50	0.54	0.28
BLEU(normalized)	0.39	0.47	0.20
Mean BERTScore	0.90	0.91	0.86
Global score (WER, BLEU and BERTScore)	1.20	1.33	0.77

Normalized scores for the diverse models

The tables above provide an overview of our results. To obtain a score that takes into account each of the three metrics, we normalized the scores (inspired by ChatGPT, using a prompt on how to calculate average scores for different metrics). In fact, we couldn't simply calculate the average because, depending on the metric used, a score close to 1 can be good (BLEU) while with another metric like WER, a score close to 1 is bad. We decided to normalize the WER scores by shifting them upward. To do this, we simply subtracted the WER score from 1 (1 – WER) to obtain a normalized score. For the BERTScore, we calculated the average of the precision scores for each sentence and consider this average to be the model's average precision. Finally, the BLEU scores obtained were expressed as percentages. To normalize them, we divided them by 100, resulting in a value between 0 and 1. We then calculated the average score for each model by summing the normalized scores for BLEU, WER, and the average BERTScore. As these results show, the GPT model achieves the best results, both in terms of the normalized overall score (1.33) and the unnormalized scores (BERTScore of 0.91, WER of 0.45, and BLEU of 47.98%). The Helsinki model follows with a normalized overall score of 1.20, BERTScore: 0.90, WER: 0.49, and BLEU: 39.91%.

Finally, the Google model has the least performance, achieving a normalized overall score of 0.77, a BERTScore of 0.86, a WER of 0.71, and a BLEU score of 20.83%.

Analysis

Overall, the results produced by all models are acceptable, although some incorrect translations are observed. The translations generated by Helsinki-NLP/opus-mt-en-fr are satisfactory and sometimes very close to the reference, as in the first sentence in English [Decorate your home for the holidays with these festive lights.] (reference: *"Décorez **votre chez-vous** pour les fêtes avec ces lumières festives"*),for which Helsinki proposes: *"Décorez **votre maison** pour les vacances avec ces lumières festives."*

The same applies to openai/gpt-oss-120b:free, which produces for the same sentence: *"Décorez **votre maison** pour les fêtes avec ces lumières festives."* A difference lies in the use of the evaluative expression *"**votre chez-vous**"* in the reference (in English, the opposition translates to "*your house*" vs "*your home")*. However, the translations diverge with respect to the English term *"holidays"*, which is rendered as *"les fêtes"* in the reference and by the GPT model, whereas Helsinki proposes *"les vacances."*

As for the google model, it fails to capture the nuance between *"lumières"* and *"étoiles"* (lights and stars), producing the translation: *"Décorer votre maison pour les fêtes avec **ces étoiles festivités.**" (Decorate your house for holidays with these festive stars).* Moreover, it renders "*festivités*" (festivities), a noun where the reference was an adjective (*festives*) in English, festive.

Overall, the translations produced by the google model are the furthest from the reference translations, as also reflected by the normalized average score across the different evaluation metrics (0.77). The model sometimes mixes English and French spelling, as illustrated in *pro**nun**ciation* (English) rather than *pro**non**ctiation/pro**non**cé* (French): *"La pro**nun**ciation est relativement facile en Italienne, car la plupart des mots sont pro**nun**ciés exactement comme ils sont écrits"* (sentence in English: Pronunciation is relatively easy in Italian since most words are pronounced exactly how they are written).

A similar issue appears in the translation of sentence 11, originally in English (*"Thirty-two split-skin grafts were studied…"*), which the model renders as: *"**Thirty-two** grafts de sèvres à sèvres ont été étudiés…"*, mixing English and French elements and producing incorrect or unnatural forms.

In addition, some word order variations may affect meaning. For example, in sentence 5 (A simple popular dinner, especially during the summer, is the Pa amb Oli: Bread with olive oil, tomato, and any available condiments such as cheese, tunafish, etc.*)* the reference translation is *"Un dîner simple et populaire…"*, whereas Helsinki proposes

*"Un simple **dîner populaire**…"*. This introduces a slight semantic nuance, as *"simple dîner populaire"* may, depending on context, carry a pejorative connotation close to "a commoner's dinner".

Furthermore, for the sentence *"*Its all-pervading power affected everyone from king to commoner.*"*, whose reference translation is *"Son pouvoir omniprésent touchait tout le monde, du roi au roturier."*, the models produce different outputs. Helsinki-NLP/opus-mt-en-fr provides an incorrect translation of *"*all-pervading power*"* (*"Son pouvoir tout-pervade…"*) and mistranslates *"commoner"* as *"communier"*. The GPT model produces a more accurate translation, closer to the reference (*"Son pouvoir omniprésent a affecté tout le monde, du roi au roturier."*). The google model correctly interprets *"all-pervading power"*, but makes an error on *"roturier"*, translating it as *"comménère."*

The fact that the google model often produces translations that are far from the reference, and sometimes even completely incorrect, can likely be explained by the nature of the prompt provided as input. Indeed, the model was simply given the instruction *"translate the following text from French to English"*, without additional information about the domain or examples, unlike the prompt provided to the GPT model, which therefore benefited from richer contextual information. This highlights the important role played by input instructions and how they can influence model outputs.

Furthermore, it can be hypothesized that the lower performance of the google model is due to the fact that, unlike Helsinki-NLP/opus-mt-en-fr, which is specifically designed for translation, google/flan-t5-large is trained on a wide range of NLP tasks, of which translation is only one and not its primary focus.

As for openai/gpt-oss-120b:free, one may assume that model size plays a central role. Indeed, this model contains 120 billion parameters, whereas google/flan-t5-large has approximately 780 million. However, these explanations remain hypotheses, although they provide interesting directions for further investigation.

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